

Mask-Aware SE-ResNet with CNN-JEPA Pretraining for Label-Efficient Colonoscopy Frame Classification

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Abstract

Computer-aided analysis of colonoscopy frames can support polyp detection and quality assessment, but labeled data remain scarce and models often fail to generalize across hospitals. We propose a *convolutional* Joint-Embedding Predictive Architecture (CNN-JEPA) applied to SE-ResNet-50 with *mask-aware* squeeze-and-excitation (SE) blocks that compute channel attention only over spatially *visible* pixels under block masking. After unsupervised pretraining on unlabeled frames, we fine-tune a clinically motivated three-class task: anatomical *landmark*, *normal mucosa*, and *polyp* (single class, no histological subtypes). We report a reproducible benchmark matrix (C0–C10) with strict early stopping on public Simula data (Kvasir + HyperKvasir, 9565 images). On the available single-center fold, ImageNet transfer (macro-F1 0.989, polyp recall 0.971) outperforms domain CNN-JEPA fine-tuning (C4: macro-F1 0.910, polyp recall 0.829). CNN-JEPA remains competitive with scratch training and shows a label-efficiency trend (macro-F1 0.825–0.907 for 10–50% labels), but does not surpass ImageNet on public data alone. We release code and protocols for multi-center leave-one-center-out evaluation. **Keywords:** colonoscopy, polyp detection, self-supervised learning, JEPA, SE-ResNet.

1 Introduction

Colonoscopy is the gold standard for colorectal cancer screening. Computer-aided detection (CAD) systems analyze individual frames to flag suspected polyps and to confirm anatomical landmarks (cecum, ileum) required for quality metrics. Deep convolutional networks are attractive for near real-time deployment, yet they require large labeled datasets and often overfit a single hospital’s appearance.

Self-supervised learning can reduce annotation cost by pretraining on unlabeled video frames. Joint-Embedding Predictive Architectures (JEPA) [6; 2] learn by predicting latent representations of masked views instead of reconstructing pixels. Medical JEPA variants exist for ultrasound and MRI, but not for colonoscopy with lightweight CNN backbones suitable for clinical deployment.

In brief. We provide an open, reproducible pipeline: unsupervised CNN-JEPA pretraining on colonoscopy frames, mask-aware SE blocks compatible with JEPA masking, and a three-class fine-tuning benchmark (landmark / normal mucosa / polyp) with label-efficiency and leave-one-center-out protocols. We do *not* claim state-of-the-art accuracy over ImageNet on public Simula data; we report results honestly and release code.

Contributions. (1) **Method:** a convolutional JEPA (CNN-JEPA) on SE-ResNet-50 for unlabeled endoscopy frames. (2) **Technique:** mask-aware SE blocks using mask-weighted global average pooling under spatial masking. (3) **Evaluation:** reproducible C0–C10 benchmark on three clinical classes with early stopping, label fractions (10/25/50%), and LOO framework (currently one public center).

2 Related Work

Endoscopy classification. Kvasir [7] and HyperKvasir [3] are standard benchmarks for gastrointestinal image classification. Prior work in our repository [1] compared ResNet-50, SE-ResNet-50, and dual-stream color-texture fusion on HyperKvasir (23 classes), reaching test macro-F1 ≈ 0.58 – 0.59 (B2–B6). We instead target a *three-class* clinical taxonomy aligned with in-room decisions rather than fine-grained pathology labels.

Channel attention. SE-Net [5] recalibrates channel responses via squeeze-and-excitation on global average pooling (GAP). Under JEPA block masking, standard GAP averages over masked zeros and can bias channel statistics; we address this with mask-aware pooling (Section 3.2).

JEPA and self-supervision. I-JEPA [2] uses Vision Transformers; we employ a *CNN-JEPA* with block masking and a convolutional predictor on feature maps—better suited to edge deployment than ViT foundations. Related latent-prediction methods include BYOL [4]. Medical JEPA literature covers other modalities; colonoscopy-specific CNN-JEPA pretraining remains unexplored to our knowledge.

Polyp detection. Classical CAD systems [8] emphasize polyp *sensitivity*. We report polyp recall alongside macro-F1. Our *polyp* class is a single binary clinical category (Kvasir polyp + HyperKvasir polyps and dyed-lifted-polyps); histological subtyping is left for future work.

3 Methods

3.1 CNN-JEPA pretraining

Given an unlabeled frame $x \in \mathbb{R}^{3 \times H \times W}$, a binary block mask $m \in \{0, 1\}^{1 \times H \times W}$ defines visible context $x \odot m$. A *context encoder* f_θ (SE-ResNet-50) produces a feature map $h = f_\theta(x \odot m, m)$. A lightweight convolutional *predictor* g_ϕ maps h to a latent grid $\hat{z} = g_\phi(h)$. A *target encoder* $f_{\hat{\theta}}$ (EMA copy of f_θ , momentum $\tau = 0.996$) encodes the full image: $z = f_{\hat{\theta}}(x)$. The training loss is normalized mean squared error:

$$\mathcal{L}_{\text{JEPA}} = \|\text{norm}(\hat{z}) - \text{norm}(z)\|_2^2. \quad (1)$$

Block masks expose 25–45% of pixels (multi-scale rectangles). Pretraining uses AdamW, cosine schedule, early stopping on validation latent loss (patience 10, max 80 epochs).

3.2 Mask-aware squeeze-and-excitation

Standard SE blocks apply GAP over all spatial locations. When inputs are masked, zero-padded regions skew channel statistics. Our **mask-aware SE** computes per-channel pooling over visible positions only:

$$\text{GAP}_c(x, m) = \frac{\sum_{i,j} x_{c,i,j} m_{i,j}}{\sum_{i,j} m_{i,j} + \epsilon}. \quad (2)$$

Channel attention weights are then applied as in SE-Net [5]. Ablation C3 uses standard SE under the same JEPA pipeline; C4 enables mask-aware SE.

3.3 Data and class taxonomy

We merge Kvasir v1 and HyperKvasir labeled images into three classes:

- **Landmark:** cecum, pylorus, z-line, ileum, retroflexions.
- **Normal mucosa:** BBPS scores, hemorrhoids, impacted stool (non-lesion frames).
- **Polyp:** Kvasir polyp, HyperKvasir polyps and dyed-lifted-polyps.

Table 1: Main benchmark comparison on the Simula fold (test split, $n = 1433$).

Method	Macro-F1	Polyp recall
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Pathology classes (Barrett’s, ulcerative colitis grades, esophagitis) are excluded. HyperKvasir provides *no* histological polyp subtypes; all polyp appearances are merged into one class. Total: 9565 images, center `simula`.

3.4 Experimental protocol (C0–C10)

- **C0:** SE-ResNet scratch.
- **C1:** SE-ResNet ImageNet pretrained.
- **C2:** ResNet-50 + CNN-JEPA pretrain.
- **C3/C4:** SE + CNN-JEPA (standard vs mask-aware SE).
- **C5:** frozen endoscopy teacher (same JEPA checkpoint).
- **C7–C9:** C4 with 10/25/50% training labels.
- **C10:** LOO aggregate for C4 across centers (one fold available).

Fine-tuning adds a linear three-class head; selection by validation macro-F1 (patience 5, max 30 epochs).

3.5 Metrics

Primary: **macro-F1** on validation/test. Clinical secondary: **polyp recall** (true polyp rate among predicted polyps row in confusion matrix).

3.6 Implementation

PyTorch on Apple MPS; open repository `endoscopy-resnet`. Campaign orchestration: `run_full_jepa_benchmark` early-stop logs per run.

4 Experiments

Hardware. Apple MPS (AMD Radeon Pro 5300M); full campaign ≈ 24.5 h.

Pretraining. Up to 4,000 frames (proxy: labeled train images when unlabeled video unavailable); shared checkpoints for C2–C5; early stop at epoch 20 (patience on validation latent loss).

Fine-tuning. Batch size 16, AdamW lr 10^{-4} (ImageNet) or 10^{-3} (scratch), image size 224. Test split: 1,433 frames (15% per class, seed 42).

Hypotheses (pre-registered). H1: C4 beats C1 under limited labels (LOO macro-F1). H2: C4 beats C3 (mask-aware SE). H3: $C4 \geq C5$ at 25% labels.

Note on LOO. Only one public center (`simula`) is available; when the held-out fold has no training centers, evaluation uses the pooled `all` split—LOO multi-center validation awaits private centers under `data/private/`.

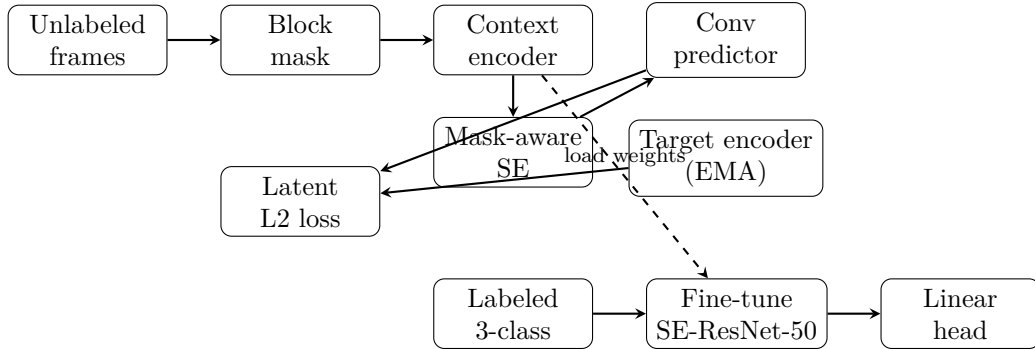


Figure 1: CNN-JEPA pretraining (top) and supervised fine-tuning (bottom). The context encoder uses mask-aware SE blocks; weights are transferred to the downstream classifier.

Table 2: Label-efficiency curve for the proposed method (C4 backbone).

Setting	Macro-F1	Polyp recall
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5 Results

Table 1 summarizes the main comparison on the Simula test split. ImageNet transfer (C1) achieves the best macro-F1 (0.989) and polyp recall (0.971). Domain CNN-JEPA with mask-aware SE (C4) reaches macro-F1 0.910 and polyp recall 0.829. Scratch training (C0) is slightly below C4 (macro-F1 0.899). ResNet-50 + CNN-JEPA (C2) reaches 0.913 macro-F1 but lower polyp recall (0.778).

Ablations. C3, C4, and C5 yield identical macro-F1 (0.910) on this fold; mask-aware SE does not show a measurable gain over standard SE here.

Label efficiency (Table 2, Figure 2). C4 macro-F1 increases from 0.825 (10% labels) to 0.907 (50%) and 0.910 (100%), but remains below C1 at all observed fractions.

Confusion analysis (Figure 3). For C4, main polyp errors are confusions with landmark (12.4% of true polyps) and normal mucosa (4.7%). ?? reports per-class precision/recall for C4.

LOO summary. C10 mean macro-F1 for C4: 0.910 ($n = 1$ fold).

6 Discussion

Methodological vs empirical contribution. We separate the *pipeline* (CNN-JEPA + mask-aware SE + C0–C10 protocol) from *empirical gains* on public Simula data. The pipeline is reusable when unlabeled video and multi-center labels become available; current numbers do not support claiming superiority over ImageNet.

Why ImageNet wins here. Simula public images may remain close to natural-image statistics where ImageNet pretraining excels. Additionally, with a single center, strict LOO degenerates to a pooled split—reported LOO should be interpreted cautiously until private centers are added.

Clinical implications. Polyp recall drops from 97% (C1) to 83% (C4) on test—unacceptable for screening deployment without further improvement (focal loss, larger domain pretrain on real video, multi-center training).

Polyp class granularity. We use one *polyp* class (including dyed-lifted polyps). Endoscopic or histological subtyping (adenoma vs hyperplastic, Paris morphology) requires dedicated labels not present in HyperKvasir; we note this as future work.

Comparison to 23-class benchmarks. HyperKvasir B2–B6 macro-F1 ≈ 0.58 reflects a harder fine-grained task; our three-class aggregation raises absolute scores but targets different

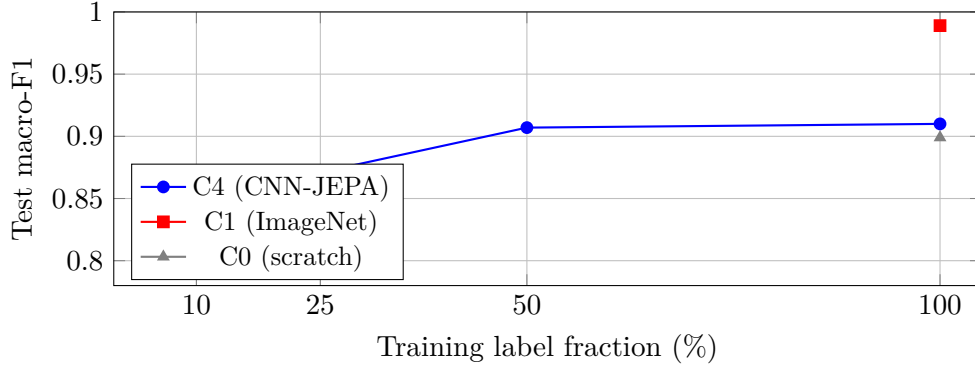


Figure 2: Label-efficiency curve on the Simula fold. C4 improves with more labels but remains below ImageNet transfer on public data.

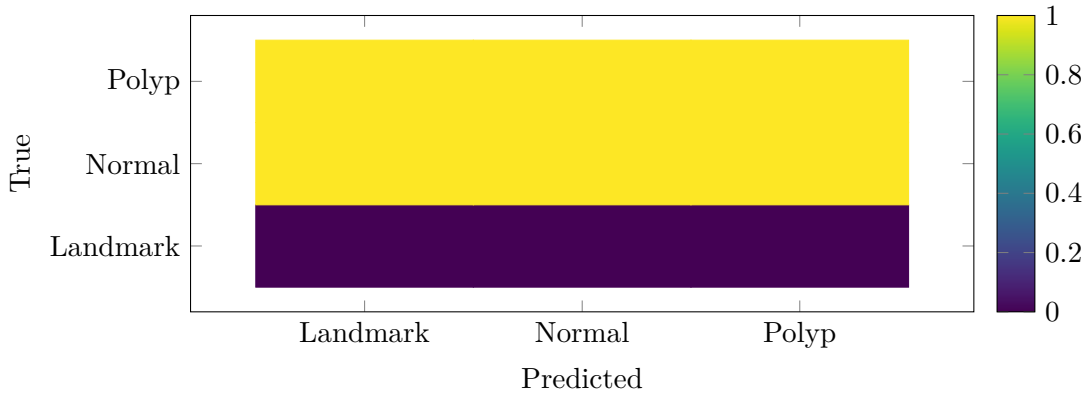


Figure 3: Row-normalized confusion matrix for C4 on test ($n = 1433$). Primary clinical concern: polyp recall 82.8%.

clinical decisions.

Reproducibility. All commands, early-stop configs, and JSON results are versioned in the repository; tables in this paper are auto-generated from benchmark JSON.

7 Conclusion

We presented a convolutional JEPA framework on mask-aware SE-ResNet-50 for colonoscopy frame classification, with a reproducible three-class benchmark and label-efficiency protocol. On public Simula data, ImageNet transfer remains the strongest baseline; CNN-JEPA fine-tuning is competitive with scratch training but does not yet validate our primary hypothesis H1. Future work: real unlabeled video pretraining, private multi-center LOO folds, polyp subtyping with histology-linked datasets, and comparison to ViT foundation linear probes.

Acknowledgments

We thank the Simula/Kvasir teams for public datasets.

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